

KALEY BRAUER

Harvard University, Center for Astrophysics | Harvard & Smithsonian
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EDUCATION

Ph.D. in Physics Massachusetts Institute of Technology Astrophysics Division; GPA: 5.0/5.0	2023
B.Sc. in Physics Brown University University honors, departmental honors, GPA: 4.0/4.0	2017

ACADEMIC POSITIONS

NSF Postdoctoral Fellow Center for Astrophysics Harvard & Smithsonian	2023 – present
US Department of Energy Graduate Fellow; Graduate Researcher Massachusetts Institute of Technology	2017 – 2023
Visiting Research Fellow Lawrence Berkeley National Laboratory	Fall 2021

HONORS & AWARDS

Research Corporation for Science Advancement Fellow	2025 – present
NSF Astronomy & Astrophysics Postdoctoral Fellowship (\$330,000)	2023 – present
Department of Energy Computational Science Graduate Fellowship (~ \$450,000)	2018 – 2022
Spot Awards, MIT, <i>in recognition of community service to School of Science</i>	2020 and 2022
NSF Graduate Research Fellowship (~ \$150,000), <i>offer declined</i>	2018
Whiteman Fellowship (~ \$100,000), MIT	2017 – 2018
R. Bruce Lindsay Award, Brown University, <i>given to senior for excellence in physics</i>	2017
Eva A. Mooar Prize, Brown University, <i>given to woman for academic excellence</i>	2017
Sigma Xi Research Honor Society	2017
Karen T. Romer Undergraduate Teaching and Research Award, Brown University	2015

PUBLICATIONS

First Author or Major Contributions

7. **Brauer, K.**, Mead, J., Ji, A. P., Wise, J. H., Bryan, G. L., Mac Low, M. M., Emerick, A., Andersson, E., Frebel, A., Cote, B. (2025). AEOS: The Impact of Pop III Initial Mass Function and Star-by-Star Models in Galaxy Simulations. *submitted to The Astrophysical Journal*. arXiv:2502.20433
6. Mead, J., **Brauer, K.**, Bryan, G. L., Mac Low, M. M., Ji, A. P., Wise, J. H., Emerick, A., Andersson, E., Frebel, A., Cote, B. (2025). AEOS: Transport of metals from minihalos following Population III stellar feedback. *The Astrophysical Journal*, 980, 62.

5. **Brauer, K.**, Emerick, A., Mead, J., Ji, A. P., Wise, J. H., Bryan, G. L., Mac Low, M. M., Cote, B., Andersson, E., Frebel, A. (2025). AEOS: Star-by-Star Cosmological Simulations of Early Chemical Enrichment and Galaxy Formation. *The Astrophysical Journal*, 980, 41.
4. **Brauer, K.**, Andales, H., Ji, A. P., Mardini, M., Frebel, A., Gomez, F. A., & O’Shea, B. W. (2022). Possibilities and Limitations of Kinematically Identifying Stars from Accreted Ultra-Faint Dwarf Galaxies. *The Astrophysical Journal*, 937, 14.
3. **Brauer, K.**, Ji, A. P., Drout, M., Frebel, A. (2021). Collapsar R-Process Yields Can Reproduce [Eu/Fe] Abundance Scatter in Metal-Poor Stars. *The Astrophysical Journal*, 915, 81.
2. **Brauer, K.**, Ji, A. P., Frebel, A., Dooley, G. A., Gomez, F. A., & O’Shea, B. W. (2019). The Origin of r-process Enhanced Metal-Poor Halo Stars In Now-Destroyed Ultra-Faint Dwarf Galaxies. *The Astrophysical Journal*, 871, 247.
1. **Brauer, K.**, Vrtillek, S. D., Peris, C., & McCollough, M. (2018). Phase-resolved spectroscopy of the low-mass X-ray binary V801 Ara. *Monthly Notices of the Royal Astronomical Society*, 478, 4894-4904.

N-th Author

4. Chiti, A., Mardini, M. K., Limberg, G., Frebel, A., Ji, A. P., Reggiani, H., Ferguson, P., Andales, H. D., **Brauer, K.**, Li, T. S., Simon, J. D. (2024). Signatures of Extragalactic First Stars in the Large Magellanic Cloud. *Nature Astronomy*, 8, 637647.
3. Ji, A. P., Naidu, R., **Brauer, K.**, Ting, Y., & Simon, J. (2022). Chemical Abundances of the Typhon Stellar Stream. *Monthly Notices of the Royal Astronomical Society*, 519, 4467-4478.
2. Mardini, M., Frebel, A., Chiti, A., Meiron, Y., **Brauer, K.**, Ou, X. (2022). Characterization of the Metal Weak Thick Disk of the Milky Way. *The Astrophysical Journal*, 936, 78.
1. Gull, M., Frebel, A., Hinojosa, K., Roederer, I. U., Ji, A. P., **Brauer, K.** (2021). R-process-rich stellar streams in the Milky Way. *The Astrophysical Journal*, 912, 52.

Non-Refereed

2. **Brauer, K.** (2021). “I’ll Finish it This Week” And Other Lies. *arXiv April Fools Paper*. arXiv:2103.16574
1. **Brauer, K.**, Ji, A., Hattori, K., Escobar, S., & Frebel, A. (2019). Kinematics of highly r-process-enhanced halo stars: Evidence for origins in now-destroyed ultra-faint dwarf galaxies. *Proceedings of the International Astronomical Union*, 14(S353), 71-74.

PRESENTATIONS

INVITED

Astronomy & Data Science Talk at Yale University	New Haven, CT, USA; May 2025
Astronomy Seminar Talk at Columbia University	New York, NY, USA; Mar 2025
Fundamental Physics Seminar Talk at Brown University	Providence, RI, USA; Mar 2025
Astronomy Seminar Talk at Lund University	Lund, Sweden; Mar 2025
Astronomy Seminar Talk at American Museum of Natural History	New York, NY, USA; Oct 2024

Nuclear Astrophysics Seminar Talk at IRENA	East Lansing, MI (remote), USA; Sep 2024
Colloquium Summer Talk at Harvard-Smithsonian CfA	Cambridge, MA, USA; July 2023
Talk at US Department of Energy CSGF Program Review	Washington, DC, USA; July 2022
High Performance Computing Talks at IHPCCS	Athens, Greece; June 2022
Colloquium Talk at University of Melbourne	Melbourne, Australia (remote); Oct 2021
Seminar Talk at Computational Research in Boston and Beyond	Boston, USA (remote); Oct 2021
Seminar Talk at Carnegie Observatories	Pasadena, CA, USA (remote); Oct 2020

CONTRIBUTED

New Data that Challenge Underlying Assumptions in Early Galaxy Evolution	Acadia, USA; Aug 2025
Tracing Cosmic Evolution with Clusters of Galaxies V	Sesto, Italy; July 2025
Talk at Galactic Frontiers II at Dartmouth	Hanover, CT, USA; Jun 2025
Talk at First Galaxies Meeting at Oxford	Oxford, UK; Apr 2025
Talk at NSF Postdoctoral Fellows Symposium 2025	Washington DC, USA; Jan 2025
Talk at IAU Symposium 395 on Stellar Populations	Paraty, Brazil; Nov 2024
Talk at Astrophysical Origins of Carbon	Tokyo, Japan; Sep 2024
Talk at First Stars VII at CCA	NYC, NY, USA; May 2024
Talk at Galaxies from Scratch 2024	Vienna, Austria; Feb 2024
Talk at 243rd Meeting of the American Astronomical Society	New Orleans, LA, USA; Jan 2024
Talk at NSF Postdoctoral Fellows Symposium 2024	New Orleans, LA, USA; Jan 2024
Talk at Flatiron CCA Galactic Frontiers	NYC, NY, USA; July 2023
Talk at IAU Symposium 377 on Early Disk Galaxy Formation	Kuala Lumpur, Malaysia; Feb 2023
Talk at 241st Meeting of the American Astronomical Society	Seattle, WA, USA; Jan 2023
Talk at JINA-CEE Frontiers in Nuclear Astrophysics Meeting	South Bend, IN, USA; May 2022
Talk at 2021 GALAH Science Meeting	Sydney, Australia (remote); June 2021
Talk at <i>Linking the Galactic and Extragalactic</i>	Wollongong, Australia (remote); Dec 2020
Talk at 235th Meeting of the American Astronomical Society	Honolulu, HI, USA; Jan 2020
Talk at IAU Symposium 353 on Galactic Dynamics	Shanghai, China; July 2019
Poster at JINA-CEE Frontiers in Nuclear Astrophysics Meeting	East Lansing, MI, USA; May 2019
Talk at JINA-CEE Frontiers in Nuclear Astrophysics Meeting	South Bend, IN, USA; May 2018
Poster at 229th Meeting of the American Astronomical Society	Grapevine, TX, USA; Jan 2017
Poster at 47th Meeting of the Division of Planetary Science	Washington, DC, USA; Nov 2015
Talk at SETI Institute's Summer Colloquium Series	Mountain View, CA, USA; Aug 2015

SKILLS

Programming	Python, C/C++, Julia, Java, HTML/CSS, SQL
High Performance Computing	OpenMP, MPI, OpenACC, experience with the National Energy Research Scientific Computing Center (NERSC)

TEACHING

8.03 Vibrations and Waves Teaching Assistant <i>Physics Department, Massachusetts Institute of Technology</i>	Spring 2023
8.S30 Stellar Archaeology Teaching Assistant <i>Physics Department, Massachusetts Institute of Technology</i>	Fall 2022
Cosmology Course Designer and Instructor <i>Spark, MIT Educational Studies Program</i>	Spring 2021
Catalyst Computer Science Instructor <i>Citizen Schools and Mass STEM Hub</i>	Spring 2019
PHYS0220/0270 Astronomy Teaching Assistant <i>Physics Department, Brown University</i>	Fall 2014 - Spring 2015
PHYS0030 Introductory Physics Workshop Assistant <i>Physics Department, Brown University</i>	Fall 2014

LEADERSHIP & SERVICE

Leader & Mentor to Postdoc Applicants <i>Astronomy Mentorship Program for Upcoming Postdocs (AMP-UP)</i>	2023 - present
Subject-matter expert reviewer in NASA peer review <i>NASA</i>	2025
Pi Day Outreach Organizer <i>Harvard University</i>	2024 - present
Graduate Women in Physics Leader <i>Massachusetts Institute of Technology</i>	2019 - 2023
High Performance Computing Mentor *only graduate student to ever serve as a mentor <i>International High Performance Computing Summer School, Athens, Greece</i>	June 2022
Reviewer for JOSS <i>Journal of Open Source Software</i>	2022 - present
President, MIT Salsa Club <i>Massachusetts Institute of Technology</i>	2018 - 2022
Physics Representative, Diversity and Inclusion Committee <i>MIT Graduate Student Council</i>	2018 - 2022
Treasurer, Students for the Exploration and Development of Space <i>Massachusetts Institute of Technology</i>	2017 - 2021
Adopt-a-Physicist Volunteer <i>Sigma Pi Sigma</i>	2018 - 2020
Designer & Observer, MIT Sidewalk Astrogazers <i>MIT Kavli Institute for Astrophysics and Space Research</i>	2017 - 2019
Head of Design Team, The Triple Helix Magazine <i>Brown University</i>	2014 - 2017

Women in Physics Co-coordinator

2016 - 2017

Brown University

Physics Show Presenter

Summer 2014

Physics & Astronomy Department, Texas A&M University